MIL-D-18873B
20 JULY 1959
SUPERSEDING
MIL-M-18873A
11 JULY 1956

MILITARY SPECIFICATION

DECK COVERING MAGNESIA AGGREGATE MIXTURE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers deck covering for use over wood floors in ammunition box cars or ammunition vans. (See 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

P-S-606 — Soap-Powder.

UU-S-48 — Sacks, Paper, Shipping.

MILITARY

MIL-D-2905 — Deck-Covering, Light-Weight, Nonslip (Abrasive-Coated Fabric and Cement).

MIL-L-15016 — Lubricating Oil, General Purpose.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer).

3. REQUIREMENTS

3.1 **Qualification.** Deck covering furnished under this specification shall be a product which has been tested without a surface dressing and has passed the qualification tests specified herein (see 6.3).

3.2 **Material.** The material used for forming the deck covering shall consist essentially of an intimate mixture of dry ingredients and a gaging solution of magnesium chloride proportioned to form a covering which can be applied with a trowel. The ingredients shall be capable of being mixed and laid to the specified thickness and shall be capable of producing wearing surfaces which are smooth, dense, and relatively free from waves, trowel marks, depressions, and other surface defects.

3.2.1 *Dry ingredients.* The dry ingredients, proportioned on a weight basis, shall be intimately premixed, and shall consist of finely

FSC 5610

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ground caustic calcined magnesia and inert aggregates such as will preclude all possibilities of sparking.

3.2.2 *Magnesium chloride.* The magnesium chloride shall be furnished in a dry flake form readily dissolvable in tap water to form the gaging solution, to which may be added a stabilizer if necessary.

3.3 **Adhesion or bonding.** The covering shall be capable of adhering to the deck or structure on which applied without the use of clips or other devices welded to the deck.

3.4 **Resistance to staining.** The covering shall be capable of being readily cleaned at frequent intervals without appreciable loss of surface or color, and shall not be adversely affected by detergents such as soap powder conforming to Specification P-S-606, when tested as specified in 4.5.2.

3.5 **Weight.** The covering after drying, shall not exceed 5.00 pounds per square foot for a thickness of ½ inch (see 4.5.3).

3.6 **Resistance to impact.** When tested as specified in 4.5.4, the covering shall show no visible signs of chipping, cracking, or detachment from the steel plate. There shall not be more than $^{1}/_{16}$ inch of permanent indentation.

3.7 **Indentation.** When tested as specified in 4.5.5, the initial indentation of the deck covering shall be not more than 1.5 percent of the actual thickness of the covering. The covering shall show no signs of cracking or becoming detached from the steel base as the result of the indentation.

3.8 **Moisture absorption.** The covering shall not absorb more than 12.0 percent of water, based on its weight at normal atmospheric conditions, when tested as specified in 4.5.6.

3.9 **Resistance to oil.** When tested as specified in 4.5.7, the deck covering shall not show more than 10.0 percent change in weight and 1.25 percent change in volume.

3.10 **Nonslip properties.** The covering shall show coefficient of friction not less than that shown in Table I when tested as specified in 4.5.8 and after 1,500 revolutions of the wear test machine specified in 4.5.9.1.

TABLE I. *Nonslip properties*

	Contacting surface conditions		
	Dry	Wet	Oily
Coefficient of static friction:			
Leather	0.30	0.50	.....
Rubber	.60	.60	0.50
Coefficient of sliding friction:			
Leather	.30	.40	.....
Rubber	.60	.60	.30

3.11 **Resistance to wear.** The covering shall show wear not to exceed 0.165 inch, when tested as specified in 4.5.9.

3.12 **Fire resistance.** The deck covering when tested in accordance with 4.5.10, shall exhibit linear damage no greater than 3 inches. The combustion plus ignition time shall not exceed 0.0 minute.

3.13 **Bond strength in shear.**

3.13.1 The initial bond strength in shear of the deck covering when bonded to wood shall be not less than 150 pounds per square inch (p.s.i.) when tested as specified in 4.5.11.

3.13.2 The initial bond strength in shear and bond strength after oxygen bomb aging and cyclic exposure of the deck covering when bonded to steel shall be not less than 400 p.s.i. when tested as specified in 4.5.11.

3.14 **Sparking resistance.** The covering shall show no evidence of sparking when tested as specified in 4.5.12.

3.15 **Electrical conductivity.** The covering shall show electrical resistance not to exceed 1 megohm when tested as specified in 4.5.13.

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3.16 Live load resistance. The covering shall show no signs of cracking or becoming detached from the steel base on three specimens when subjected to a 3,000 pound roller load and shall show an average indentation not greater than 0.050 inch when tested as specified in 4.5.14.

3.17 Nailability. The covering shall show no signs of chipping, cracking, or detachment when tested as specified in 4.5.15.

3.18 Ease of repair. The covering shall be easily repairable and shall have the ability to be featheredged, showing good bond between cured and new material when tested as specified in 4.5.16.

3.19 Surface. The covering shall present a surface that is smooth and will permit the easy removal of scattered powder grains when tested as specified in 4.5.17.

3.20 Directions for application. Each container shall be clearly labeled with instructions for application of the coating.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Qualification tests at a Government laboratory. Qualification tests shall be conducted at the Material Laboratory, New York Naval Shipyard, Naval Base, Brooklyn 1, New York. These tests shall consist of the tests specified in 4.5 (see 6.3). Each sample shall consist of sufficient premixed dry ingredients (finish coat) and magnesium chloride flakes to provide for the application of 27 square feet, approximately ½ inch thick.

4.3 Sampling.

4.3.1 *Lot.* For purposes of sampling, a lot shall consist of all unmixed ingredients (dry ingredients and magnesium chloride flakes) offered for delivery at one time.

4.3.2 *Sampling for inspection of filled containers at the place of manufacture.* A random sample of filled containers shall be selected from each lot offered for Government inspection in accordance with Standard MIL–STD–105 at inspection level I and acceptable quality level = 2.5 percent defective to verify compliance with this specification in regard to fill, closure, and marking. Each type of container shall be sampled separately.

4.3.3 *Sampling procedure for lot acceptance at a Government laboratory.* A random sample of unmixed ingredients shall be selected by the inspector from each inspection lot in accordance with Table II. Each sample shall consist of sufficient premixed dry ingredients (finish coat) and magnesium chloride flakes to provide for the application of 9 square feet, approximately ½ inch thick. The samples so selected shall be forwarded to the Material Laboratory, New York Naval Shipyard, Naval Base, Brooklyn 1, New York.

TABLE II. *Sampling for acceptance inspection*

Lot size in pounds	Number of samples
500 and under	None
501 to 1,500	2
1,501 to 4,000	3
4,001 to 8,000	4
8,001 to 15,000	5
15,001 to 25,000	6
25,001 to 40,000	7
40,001 to 60,000	8
60,001 to 90,000	9
90,001 to 120,000	10
120,001 to 200,000	12
200,001 to 300,000	14
300,001 and over	16

4.4 Inspection and tests.

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4.4.1 *Inspection of filled containers at the place of manufacture.* Each sample filled container selected as specified in 4.3.2 shall be examined for defects of construction of the container and the closure, for evidence of leakage, and for unsatisfactory markings; each filled container shall also be weighed to determine the amount of contents. Any filled container in the sample having one or more defects, or under required fill, shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of Standard MIL–STD–105, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for inspection provided the contractor has removed or repaired all nonconforming filled containers.

4.4.2 *Lot acceptance tests at a Government laboratory.* The dry ingredients and the magnesium chloride flakes selected in accordance with 4.3.3 shall be mixed (separately for each container) in accordance with the manufacturer's instructions, and subjected separately to the tests specified in 4.5.3, 4.5.4, 4.5.5, 4.5.9, 4.5.12, and 4.5.14. If any sample is found to be not in conformance with this specification, the entire lot shall be rejected. A rejected lot may be resubmitted for Government inspection, but only after the manufacturer, subsequent to the first Government inspection, has reworked the whole lot, subjected it to the test or tests wherein failure occurred, and removed all nonconforming material.

4.5 Test procedures.

4.5.1 *Preparation of specimens.* Specimens of the sizes as required for the following tests shall be made on clean ⅛ inch thick mild steel plates, unless otherwise specified. The plates shall be primed with a brush coat of cement conforming to Specification MIL–D–2905 and allowed to dry for a minimum of 2 hours. The covering shall be prepared in accordance with the manufacturer's instructions, using a gaging solution of 21.5 ± 0.5° Baume, and trowelled on the primed steel plates to a thickness of approximately ½ inch, unless otherwise specified, by means of templates. The specimens shall then be cured for 96 hours at room temperature before conducting tests. Except as specified herein, all tests shall be conducted at a temperature of 70° to 75° Fahrenheit (F.) and a relative humidity of 50 ± 2 percent.

4.5.2 *Resistance to staining.* A specimen 6 by 2 inches shall be immersed for 1 hour in the solution specified herein so that a 3 inch section of the specimen is immersed in the solution and the upper half remains unexposed for purposes of comparison. The liquid solution shall be made by dissolving soap powder conforming to Specification P–S–606 in tepid water to form a ½ of 1 percent solution, then adding lamp black until the solution is dark gray. After immersion, the specimen shall be rinsed for 1 minute with tap water at 140° F. and dried with an air jet. The specimen shall then be examined for softening, stains, streaking, or loss of color.

4.5.3 *Weight.* The deck covering shall be applied to three 6 inch square steel plates which have been previously measured and weighed. When the material has set for 96 hours, the three test specimens, including the steel plates shall each be weighed to the nearest 0.1 gram. The length and width shall be measured to the nearest 0.01 inch, and the thickness to the nearest 0.001 inch. The thickness shall be determined by measuring the steel plates with and without the coating, at 16 equally distributed points on the specimen, by means of a dial thickness gage and a template. The difference in thickness of the steel plate and the coated steel plate shall be averaged to determine the thickness of the material. From the difference between the weights of the coated steel and the uncoated steel, the actual weight of the material shall be determined and then computed in pounds per square foot per ½ inch thickness. The final computed weight shall be the average of the three specimens.

4.5.4 *Resistance to impact.* Two specimens 6 inches square shall be used, including the full thickness of the deck surfacing and the

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steel base. Each specimen shall be tested separately while being firmly held on a solid horizontal base. A 2-pound ball shall be dropped vertically from a height of 8 feet so that the impact will be at the center of the specimen. Each specimen shall be subjected to two impacts of the ball.

4.5.5 *Indentation.* One specimen 6 inches square shall be used, including the full thickness of the deck covering and steel base to which it was applied. Three indentations shall be made on the specimen and the results averaged. The center of each indentation shall be not less than 1½ inches from the edge of the specimen, and not less than 2½ inches from the center of the adjoining indentation.

4.5.5.1 A load of 2,000 pounds shall be applied on the deck covering for 30 minutes by means of a circular indentor with a cross-sectional area of 1 square inch and a radius of 1/64 inch on the perimeter of the indenting flat face. The thickness of the deck covering shall be measured with a micrometer dial gage using a 4 ounce weight and ¼ inch diameter flat foot on both sides of the specimen before and after the 2,000 pound load has been applied, and the thickness of the steel base shall be subtracted in each case.

4.4.5.2 The indentation shall be taken as the percent change in the thickness of the material.

4.5.6 *Moisture absorption.* Three specimens 2 inches square by ¼-inch thick shall be prepared by applying the deck surfacing to oiled surfaces of steel plate, in such a manner that upon drying the specimens will not adhere to the plates. Each specimen without the steel backing plate shall be weighed dry, dipped into tap water at room temperature, lightly wiped on all surfaces with a paper towel and again weighed to the nearest 0.01 gm. Immediately after weighing, the specimen shall again be immersed in the tap water for 24 hours, lightly wiped and again weighed. The percent gain in moisture shall be based on the weight of the dry specimen and the difference between the weight after 24 hour immersion and the weight after dipping and wiping.

4.5.7 *Resistance to oil.*

4.5.7.1 *Test fluid.* The test fluid shall be oil conforming to Symbol 3042 of Specification MIL–L–15016.

4.5.7.2 *Procedure.* Three specimens, 2 inches square by ¼ inch thick, shall be prepared as specified in 4.5.6. Each specimen without the steel backing plate shall be weighed dry, dipped into the test fluid, and lightly wiped on all surfaces with a paper towel. The specimen shall then again be weighed in air, and also while totally immersed in tap water. Immediately after weighing, the specimen shall be immersed for 24 hours in the test fluid, lightly wiped and again weighed in air and while totally immersed in tap water. The percent gain in weight shall be based on the weight of the dry specimen and the difference between the weight after dipping and wiping and the weight after 24 hours immersion. The percent change in volume shall be based on the weight of the specimen immersed in water, before and after 24 hours immersion in the test fluid.

4.5.8 *Nonslip properties.* The coefficient of friction of the deck covering against leather and rubber respectively, shall be determined. The leather shall be oak-tanned sole leather which has been sanded smooth with grade 0 garnet paper. The rubber shall be a vulcanized compound with a hardness range of 60 to 80 Durometer "A". Tests shall be made with the contact surface dry, wet with a solution of 4 percent sodium chloride in water and oiled, using oil conforming to Symbol 3042 of Specification MIL–L–15016. A load of 33 pounds shall be applied uniformly over the 2 by 4 inch specimen. The determination of static and sliding friction may be made by the inclined plane method or by determining the pull on a spring balance required to start the test piece from rest and then to maintain a constant velocity. The factor of friction shall be the average of four readings.

4.5.9 *Resistance to wear.* Specimens 3 inches long by 2 inches wide shall be prepared so that an area of approximately 6 square inches is presented to the abrasive in the machine

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specified in 4.5.9.1. The abrasive grit shall be No. 80 aluminum oxide. Before wear-testing, the thickness shall be measured at 12 equally distributed points on the specimen by means of a dial thickness gage and a template. The specimen shall then be subjected to 1,500 revolutions of the wear test machine, and the thickness obtained at the same points previously measured. The difference in thickness shall be averaged to determine the thickness of material worn away by the abrasive.

4.5.9.1 *Wear-test machine.* The wear-test machine shown on figures 1 and 2 shall have the following essential mechanical characteristics:

(a) Mild steel, abrasion disk, 14⅛ inches in diameter and ½ inch thick, revolved clockwise at a constant speed of 23.5 revolutions per minute (r.p.m.).

(b) Specimen holder, 2 by 4 inches, attached to a ⅝ inch diameter shaft by means of a universal coupling. The distance between centers of the disk and the specimen holder shaft is 5.1 inches. A specimen of decking material is secured in the specimen holder by means of clamps at each end of the holder. The holder revolves clockwise at a constant speed of 32.5 r.p.m.

(c) Hardened tool set cam attached to the specimen holder shaft by means of set screws. In each ½ revolution of the shaft, the specimen is lifted a distance of 1/16 inch and then dropped back on disk specified in 4.5.9.1 (a) under a 10 pound weight. Change in the lift due to wear of the specimen is compensated by adjustment of the cam at regular intervals, so that the lift is maintained at a practically constant distance.

(d) Weight 10 pounds, the maximum found suitable for use with this test machine.

(e) Distributor for abrasive grits. The grits flow by gravity from the hopper specified in 4.5.9.1 (f) through a 13/64 inch inside diameter tube and into the center of the distribution chamber. An impeller in the chamber provides an evenly distributed feed of the grits through eight 3/64 inch diameter holes centered on a 1⅝ inch diameter pitch circle. A uniform distribution of grits falls to disk specified in 4.5.9.1 (a) in the path of the test specimen. The distributor is revolved at a constant speed of 9 r.p.m.

(f) Hopper for feeding abrasive grits.

(g) Bevel drive gear.

(h) Predetermining revolution counter and switch.

(i) One-third horsepower compound wound direct current motor.

(j) Hopper for collecting used abrasive grit.

(k) Steel frame.

Details of the distributor for the abrasive grit, the cam, and specimen holder are shown on figures 3 and 4, respectively.

4.5.9.2 *Procedure.* The steel disk specified in 4.5.9.1 (a) revolves while the weighted specimen of deck surfacing in holder specified in 4.5.9.1 (b) is lifted and dropped by a cam while it is being revolved. When the specimen is dropped through motion of the cam, it falls a distance of 1/16 inch and meets disk (see 4.5.9.1 (a)) with impact simulating the fall of a foot on the deck surface. Foot action is further simulated in that the revolving specimen contacts the disk with a sliding and twisting motion until lifted clear by the cam. Continuous rotation of the specimen introduces the element of wear in all directions on the decking material. Free abrasive grits of No. 80 aluminum oxide are fed at a steady rate from the hopper (see 4.5.9.1 (f)) and distributed uniformly on the steel disk in the path of the specimen. The revolving motions of the steel disk and the specimen keep the abrasive grits circulating to the outer side of the path of the specimen. The grits fall from the edge of the steel disk into hopper (see 4.5.9.1 (j)), from which they are collected and then screened to eliminate undesired particles before replacement in the upper feeding hopper (see 4.5.9.1 (f)).

4.5.10 *Fire resistance test.*

4.5.10.1 *Apparatus.* The fire-resistance of the deck covering shall be determined by an apparatus as shown on figure 5, which consists of communicating horizontal and vertical

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flues constructed of asbestos board sheets supported on an angle-iron frame, with the exception of the horizontal bottom plate, which is steel. The enclosure shall be 36 inches long overall; the flue shall be 8 inches wide by 6 inches high. The bottom plate shall be cut away 3 inches from the firing end to provide clearance for the flame of four open blast gas burners. Details of the fire-resistant apparatus are shown on figure 6. Details of the burner supply manifold and the open blast burner nozzle are shown on figure 7.

4.5.10.2 *Procedure.* The deck covering shall be prepared in the usual manner on a ⅛ inch mild steel plate 31½ inches long by 7 inches wide, with longitudinal flanges for mounting in the horizontal flue. The plate shall be secured in the flue, 2 inches above the bottom plate and with the inner end spaced 3 inches from the flue end, to permit hot gases passing beneath the test plate to be vented through the vertical flue. The four open blast burners shall be located side by side and parallel to the front end of the test plate, on 1¾ inch centers, equidistant from each side of the flue. The center of the burners shall be located 4 inches below the bottom surface of the test plate, and shall be ½ inch in front of the firing end of the test plate. The gas used shall be commercial propane having a heating value of 2550 British thermal units (B.t.u.) per cubic foot at a temperature of 60° F. and atmospheric pressure of 30 inches of mercury. The gas and air shall be supplied to a common manifold at a rate of 9.6 and 150 cubic feet per hour respectively, when referred to standard conditions of 60° F. and 30 inches of mercury. The burner flames shall be applied to the test plate for 4 minutes and then immediately removed. A draft of 0.06 (plus 0.01 minus 0.00) inch of water shall be maintained in the flue of the apparatus for the duration of the test.

4.5.10.2.1 Combustion plus ignition time, and linear damage shall be noted. These quantities are defined as follows:

(a) *Combustion plug ignition time.* The time taken from the initial application of burner flames until all flaming of the specimen ceases.

(b) *Linear damage.* The length of specimen permanently damaged by burning, charring, blistering, or embrittlement. This measurement shall be taken after all flaming ceases.

4.5.11 *Bond strength in shear.* Six specimens shall be prepared by applying the deck covering in an area 2 inches square to 2 by 6 inches ¼ inch thick yellow pine in a thickness of ¼ inch by means of a template, such that 1 inch of the wood base is exposed at one end, and 3 inches exposed at the opposite end. Eighteen similar specimens shall be prepared on 2 by 6 inches by ⅛ inch thick mild steel plates. Specimens shall be tested by measuring the load required to shear the 2 inch square area of deck surfacing from the wood base or steel plate by means of a shear test jig as shown on figure 8. All specimens after preparation shall be allowed to set for 96 hours. Six specimens of each of those mounted on steel and wood shall then be tested to determine the initial bond strength in shear. Six specimens of those mounted on steel shall be tested after aging in an oxygen bomb for 96 hours under a pressure of 300 p.s.i. and temperature of 70° centigrade (C.) (158° F.). The six remaining specimens mounted on steel shall be tested after being subjected to the cyclic exposure specified in 4.5.11.1.

4.5.11.1 *Cyclic exposure.* The six specimens mounted on steel shall be immersed in a solution of 4 percent sodium chloride water, under a pressure of 8 p.s.i. for 48 hours. Immediately after immersion, the specimens shall be subjected to two complete cycles of alternate exposure to a temperature of 0° ± 5° F. for 24 hours, followed by a temperature of 120° ± 5° F. for 24 hours. After this cyclic exposure the specimens shall be subjected to the shear test as specified in 4.5.11.

4.5.12 *Sparking resistance.* The three specimens used for the weight determination shall be tested for sparking resistance by abrading the specimens with a rotating stiff wire wheel attached to an electric drill and by striking the specimens glancing blows with a steel hammer. These tests shall be run in a darkened

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room and the presence or absence of sparks shall be noted.

4.5.13 *Electrical conductivity.* The deck surfacing shall be applied to two 1-foot square mild steel plates in a 1-inch thickness. The electrical resistance of the deck covering shall be determined as follows:

(a) A brass electrode shall be used for the electrical resistance measurements. The electrode shall be 2½ inches in diameter, weigh 5 pounds and have a rubber pad base of 55 durometer hardness cemented to one face. The contact surface of the electrode shall be covered with a sheet of aluminum foil 0.003 inch in thickness, held to the sides of the electrode with elastic bands.

(b) The electrical resistance shall be measured between two points, one directly at the center of the top surface of the decking material where the electrode is placed and the other at one corner of the steel base plate.

(c) The instrument used shall be an insulation tester of the true ohm-meter type, having a mechanically operated generator with an output of 500 volts d.c. and a short circuit current of 5 milliamperes.

4.5.14 *Live load resistance.* Three specimens 6 inches wide by 12 inches long and 1 inch thick shall be prepared on steel bases, mounted on the live load apparatus and each specimen subjected to the following loading procedure:

(a) A roller preload of 400 pounds shall be applied and initial thickness readings taken at four equally spaced positions of the roller as it passes over the specimen.

(b) A roller load of 1000 pounds shall then be applied and the specimens subjected to 25 cycles of reciprocation under this load.

(c) The roller load shall then be increased in 500 pound increments 25 cycles applied at each loading, until a maximum roller load of 3000 pounds is attained.

(d) At the end of the loading procedure the final thickness of each specimen shall be measured at the four equally spaced positions with the roller load reset at 400 pounds. The average decrease of thickness for each specimen is then determined by the difference between the initial and final thickness readings.

4.5.14.1 *Live load apparatus.* The live load apparatus shall consist essentially of a reciprocating table that carries a specimen of the material under test, and a roller that is brought to bear against the specimen by a pneumatic cylinder having a piston diameter of 8 inches. The axis of the roller shall be parallel to the plane of the specimen and perpendicular to the specimen's line of motion, so that the roller rotates as the specimen reciprocates under it. The specimen table shall oscillate at approximately 9 cycles per minute through a stroke of 5 inches. The roller shall be of brass, 4 inches in diameter, 3⅛ inches wide, with a ¼ inch radius at both ends. The roller contacting force shall be controlled by varying the pressure in the loading cylinder.

4.5.15 *Nailability.* The two specimens used for the electrical conductivity determination shall be tested for nailability by driving eightpenny nails 1 inch in from the edge of the specimens and inspecting for signs of chipping, cracking, or detachment of the surfacing from the steel base.

4.5.16 *Ease of repair.* The two specimens used for the electrical conductivity determination shall be tested for ease of repair by gouging out areas of various sizes to a minimum depth of ⅛ inch with a hammer and a ½ inch chisel. These areas shall then be patched, noting the ease of repair and the material's ability to featheredge as well as the bond between the patch and the original surfacing.

4.5.17 *Surface.* Specimens prepared for test shall be visually examined for surface defects that might prevent the removal of scattered powder grains.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Not applicable.

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5.2 **Packing.**

5.2.1 *Level A.* The deck covering dry ingredients and magnesium chloride flakes shall be packed in 50 or 75-pound quantities in multiwall shipping sacks conforming to construction 7X of Specification UU–S–48.

5.2.2 *Level B.* The deck covering dry ingredients and magnesium chloride flakes shall be packed in 50 or 75-pound quantities in multiwall shipping sacks conforming to construction 5W or 7W of Specification UU–S–48.

5.2.3 *Level C.* The material shall be packed for domestic shipment in accordance with the manufacturer's commercial practice. The packing shall be adequate to afford protection from damage during direct shipment from supply source, to the first receiving activity for immediate use. The shipping container shall comply with carrier's rules and regulations applicable to the mode of transportation.

5.3 **Marking.** In addition to any special marking required in the contract or order, marking for shipment shall be in accordance with Standard MIL–STD–129.

6. NOTES

6.1 **Intended use.** Deck covering covered by this specification is intended for use over wood floors in ammunition box cars or ammunition vans. It should not be used for areas or locations where it is subject to continual wetting. It is for single coat application.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Level of packing required (see 5.2).

(c) Special marking, if required (see 5.3).

6.2.1 *Criteria for use of proper level of preservation, packaging, and packing* shall be as follows:

For Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For Level B. This level shall be used only when it is definitely known that the item will be held in covered storage in overseas locations for 6 months or less or in domestic locations for an indefinite period.

For Level C. This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

6.3 With respect to products requiring qualification, awards will be made only for such products as have, prior to the bid opening date, been tested and approved for inclusion in Qualified Product List QPL 18873, whether or not such products have actually been so listed by that date. Manufacturers are urged to communicate with the Chief of the Bureau of Ordnance, Department of the Navy, Washington 25, D. C., and arrange to have the products that they propose to offer tested for qualification.

6.4 Deck covering dry ingredients and magnesium chloride flakes should be purchased by weight.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Corps of Engineers
Navy—Bureau of Ordnance

Preparing activity:
Navy—Bureau of Ordnance

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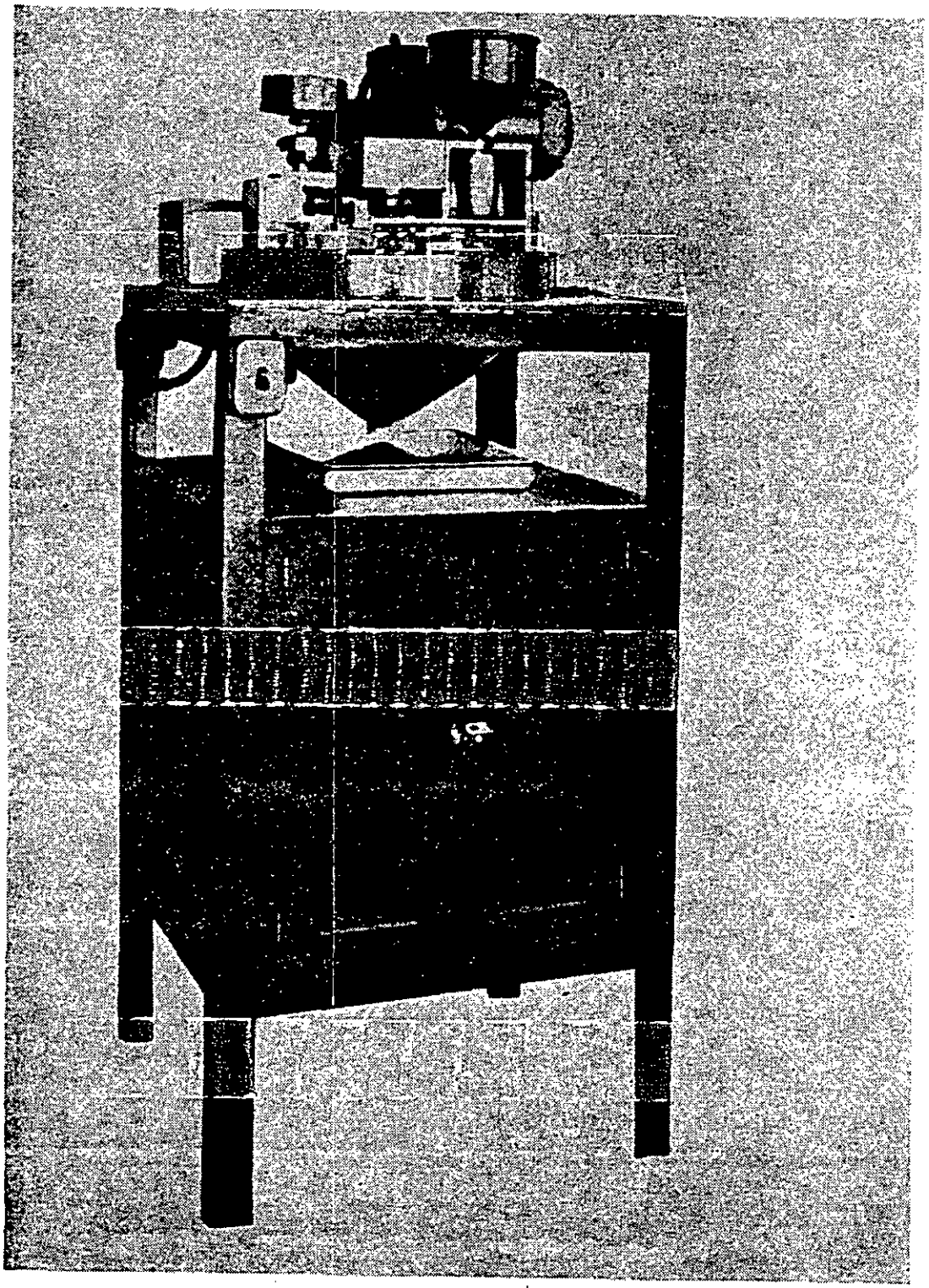


FIGURE 1. *Wear test machine.*

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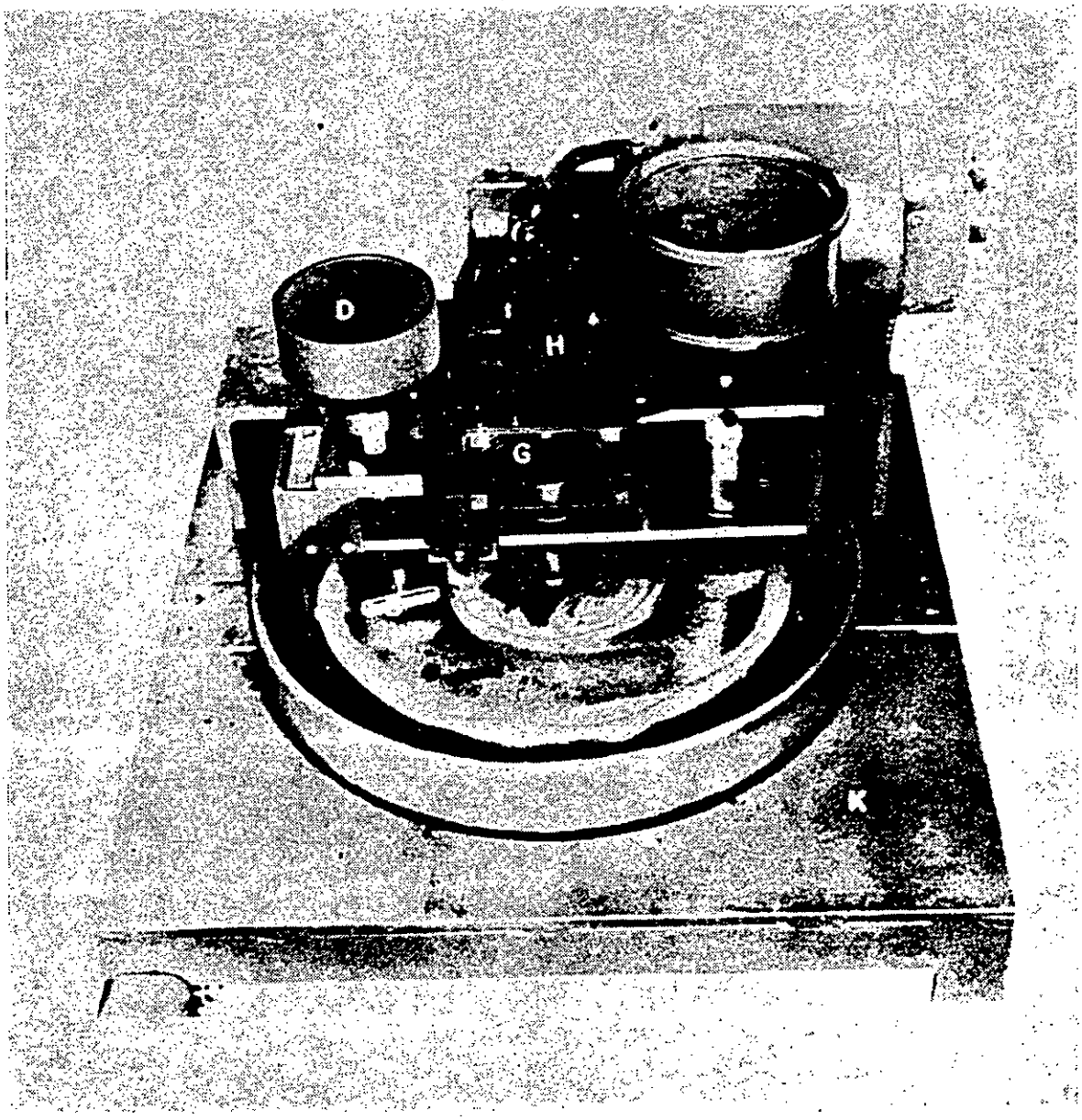


FIGURE 2. *Detail view of wear test machine.*

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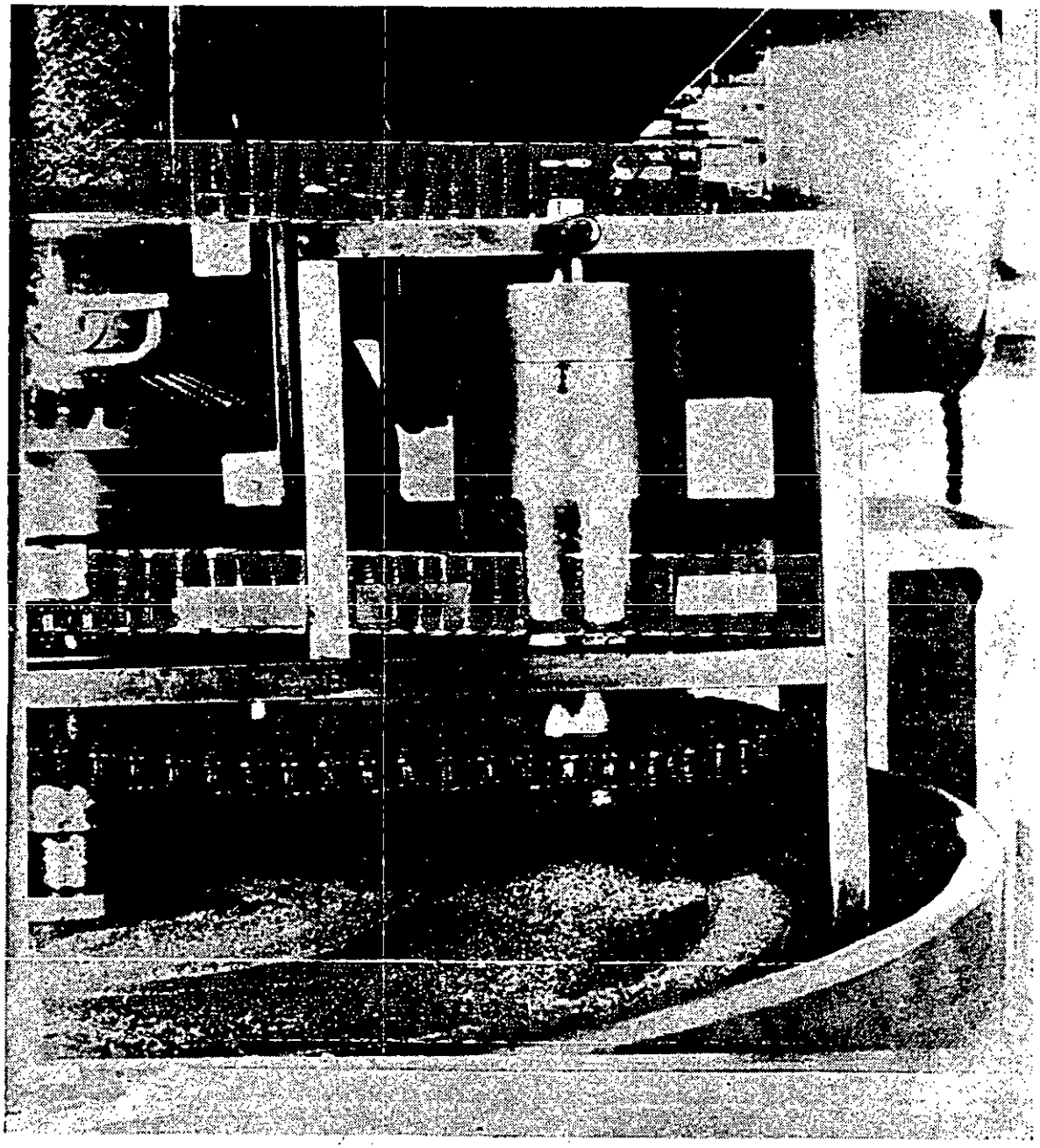
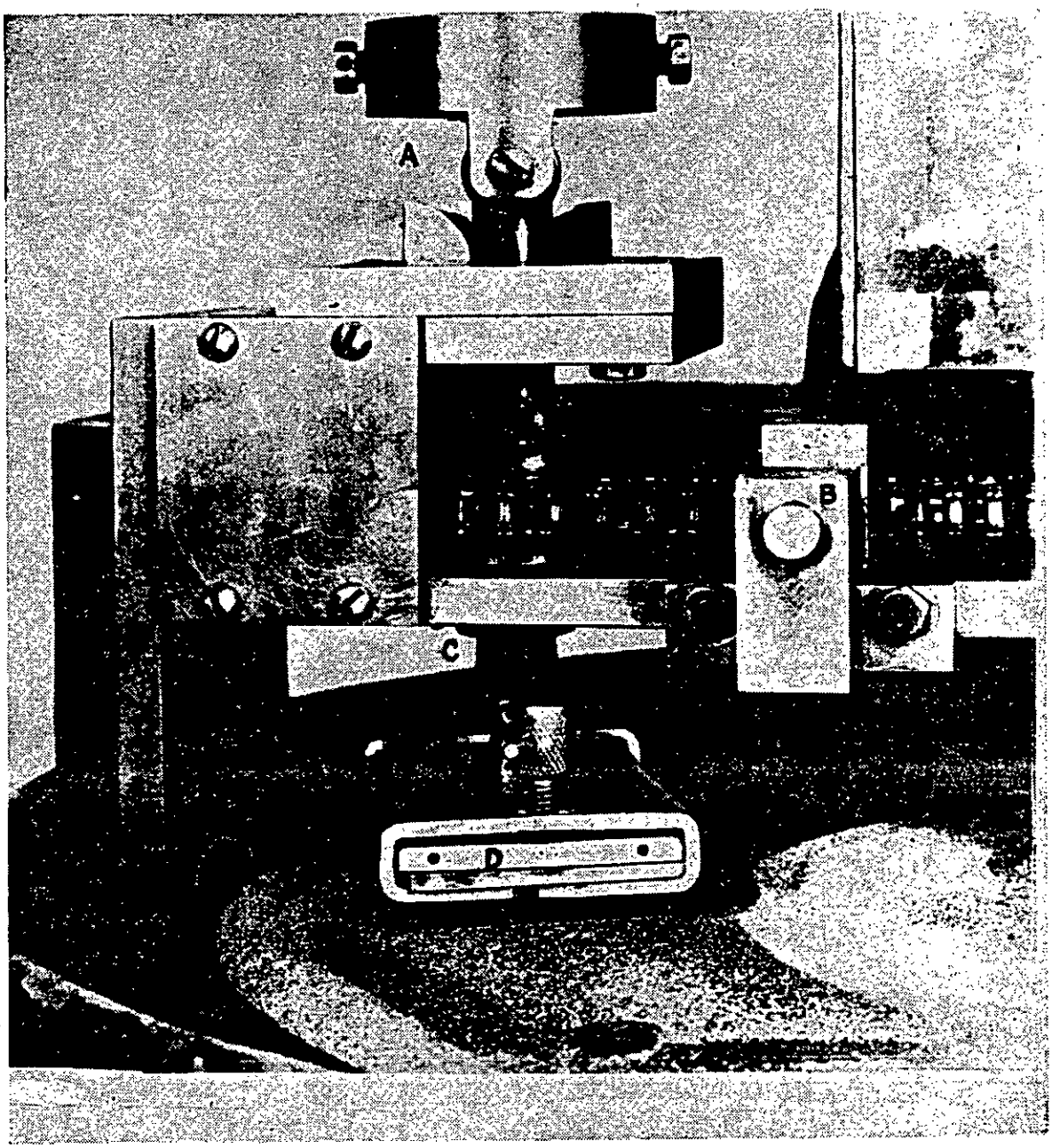


FIGURE 3. *Detail of distributor of abrasive grits.*

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A — Cam.
B — Chain idler.
C — Universal joint.
D — Specimen holder.

FIGURE 4. *Detail of cam and specimen holder.*

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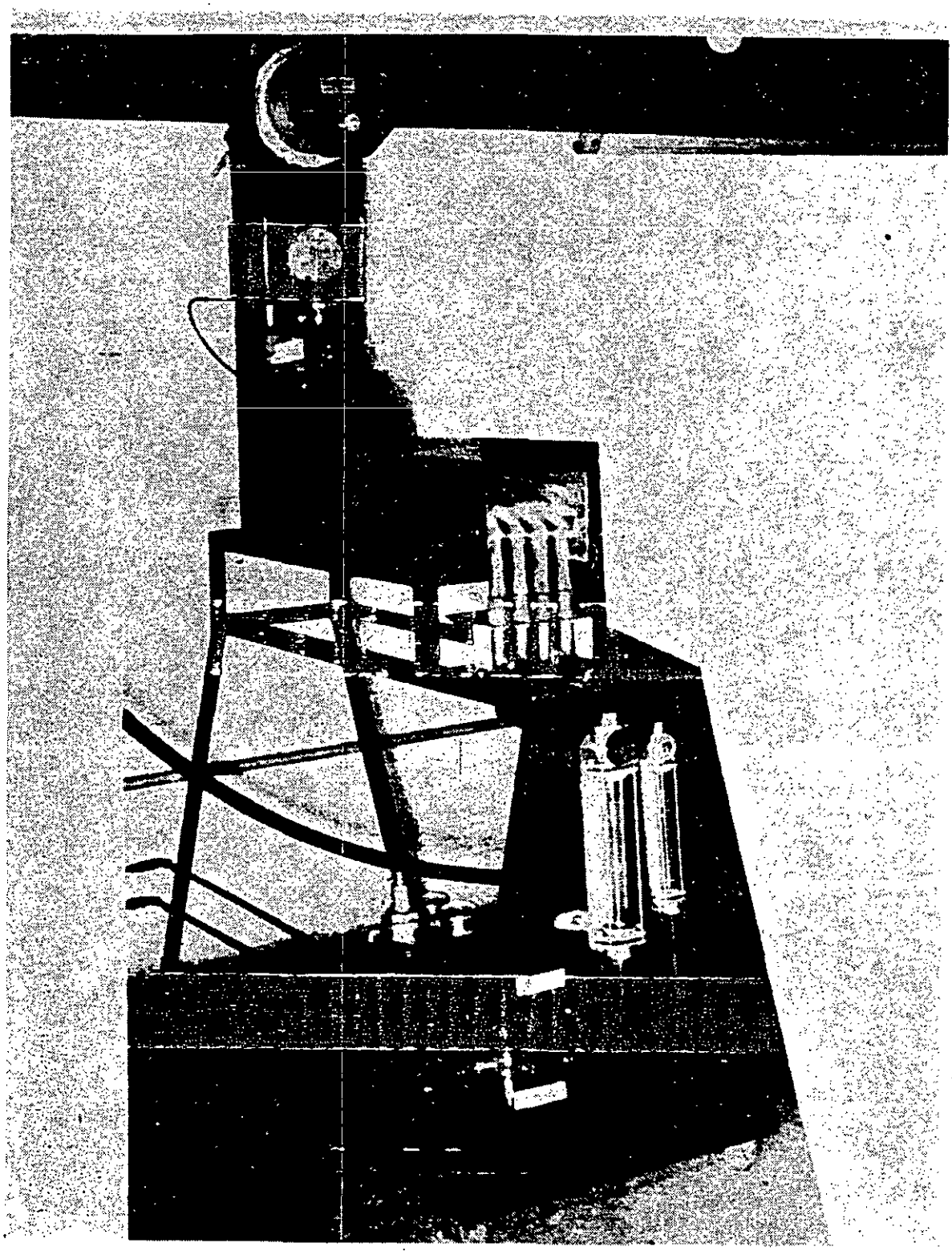


FIGURE 5. *Fire-resistance test apparatus.*

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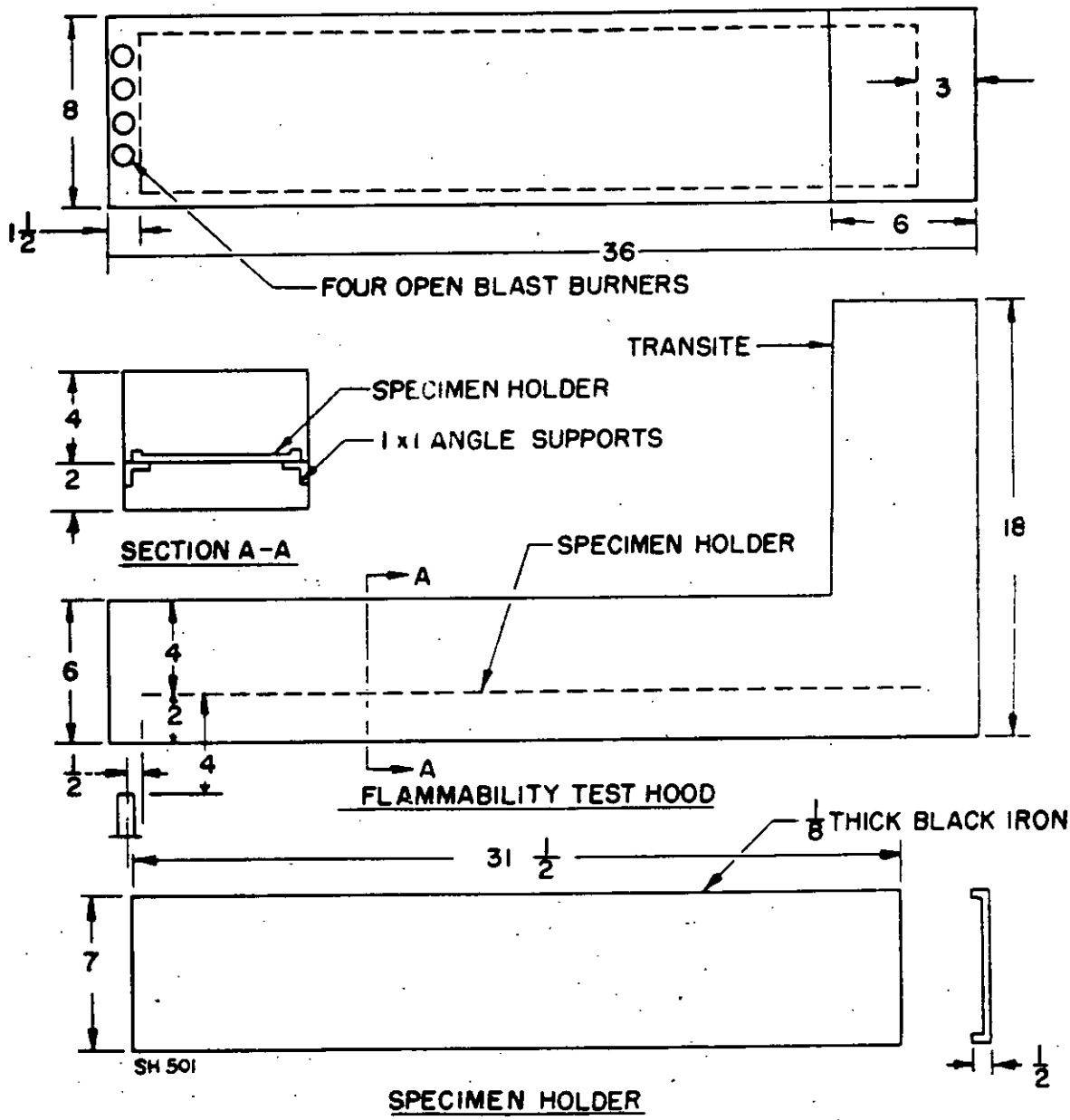


FIGURE 6. *Fire-resistance test apparatus details.*

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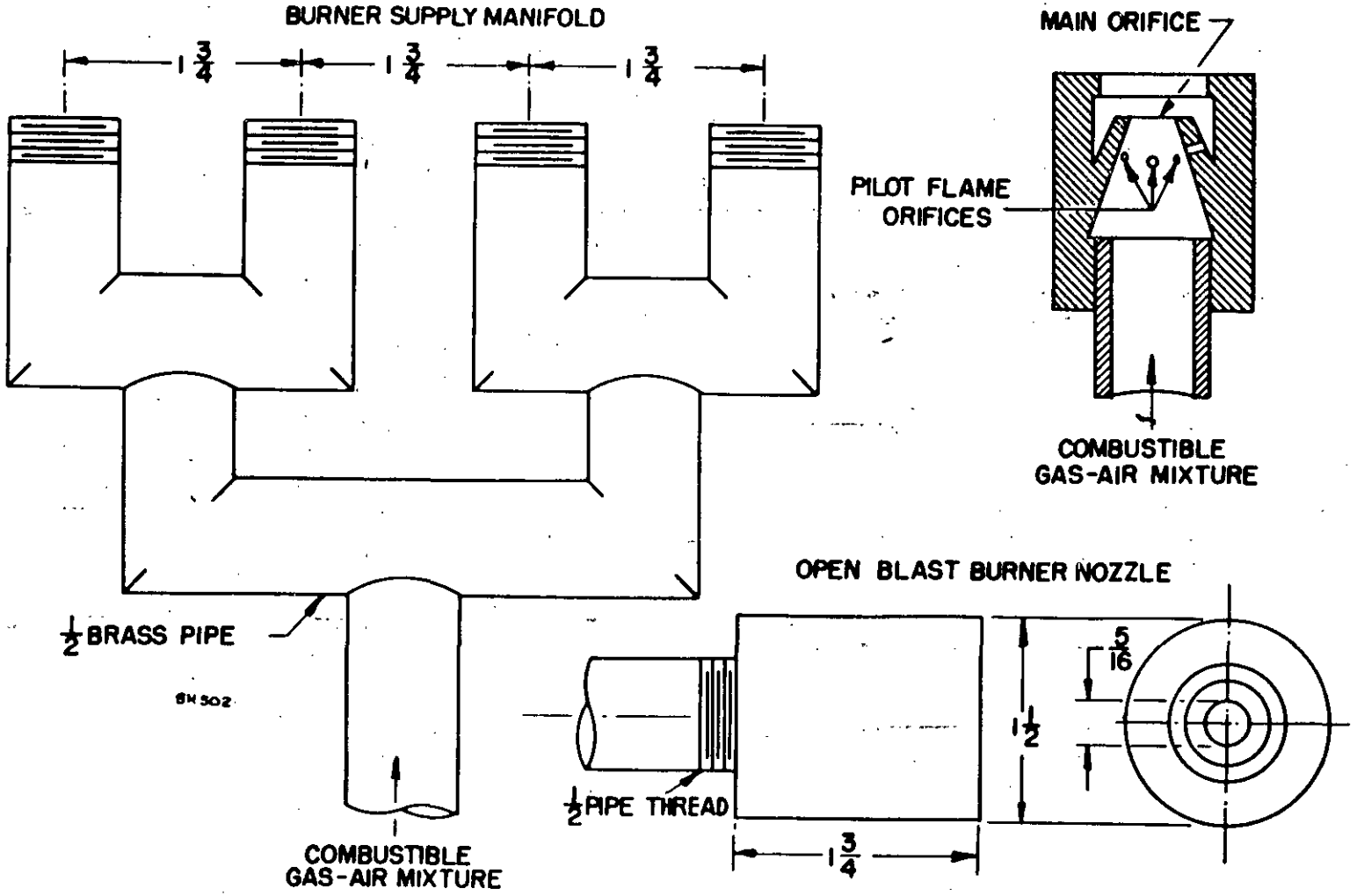


FIGURE 7. *Burner supply manifold and open blast burner nozzle details.*

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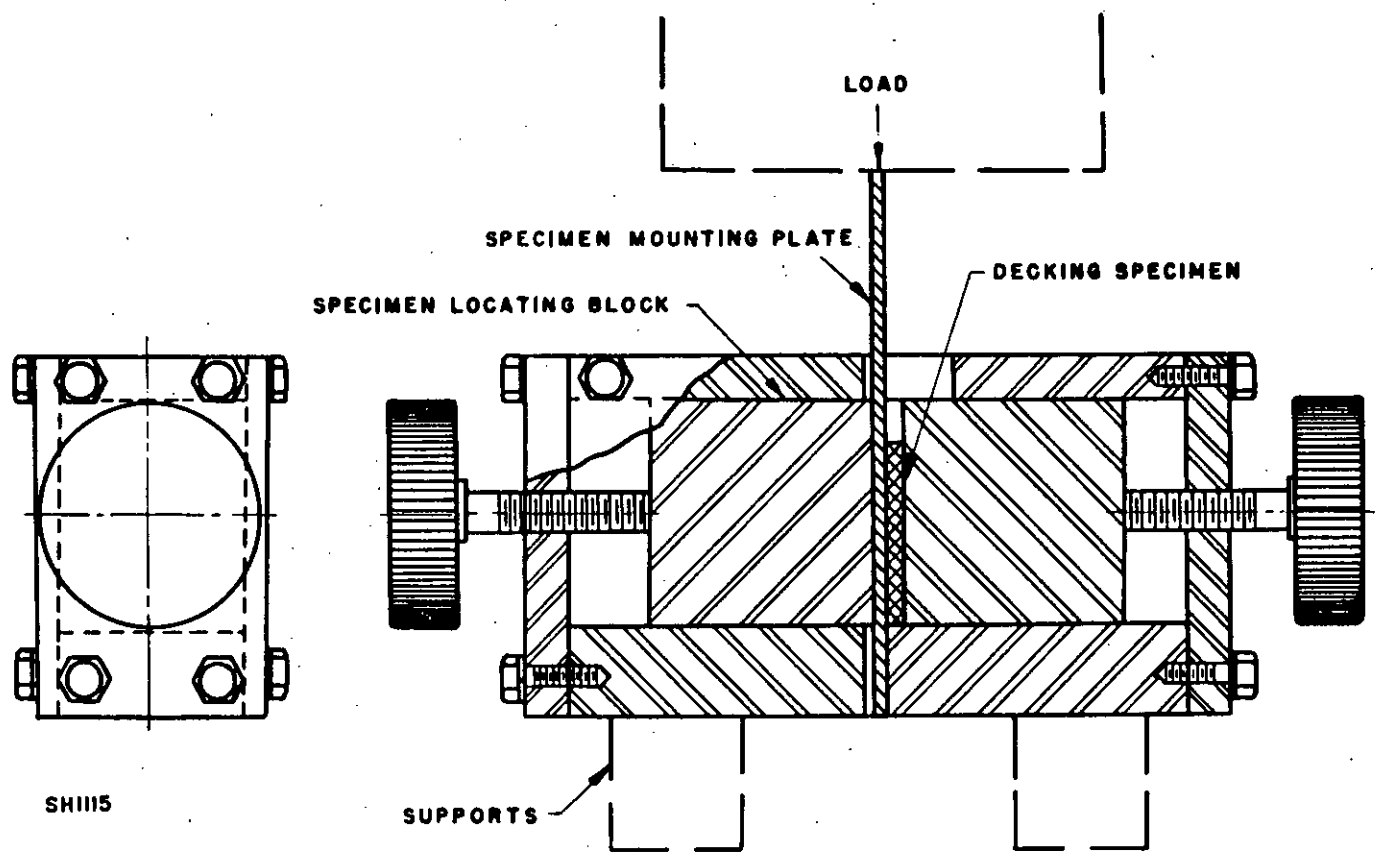


FIGURE 8. *Shear test jig for hard-setting deck covering materials.*

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a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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